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About the Signal Definition Explorer

About the Signal Definition Explorer

The Signal Definition Explorer (SDE) is a GUI tool for browsing and editing Teradyne Signal Definition (TSD) files and managing libraries of TSD files. SDE includes functions to manage the waveform data in TSD files and to move or copy TSD files to test program development folders outside of the TSD libraries.

Although it was primarily developed for use with the Modulation Toolkit option, you can use SDE to create, store, manage, and apply waveform data independently of the Modulation Toolkit. SDE operates as a standalone tool. It is installed with IG-XL and does not require a separate license. Using SDE, you can:

- Access the Teradyne-supplied Signal Library on your system or in your environment
- Create your own user library to organize and manage TSD files for your test programs
- Copy and move TSD files to your development folders. SDE automatically copies or moves any VSA setup or utility files referenced by a TSD file.
- Select TSD files directly in a view tree or by defining a file selection filter using TSD file properties
- Edit TSD files as required to create the specific files needed for your test program
- Convert IQ waveform data to IF
- Split a TSD file containing IQ data into separate files containing I and Q data
- Import signal definition data created by EDA applications into TSD files
- View signal waveform samples using WaveScope, VectorScope, or VSA (license required)

Signal Definition Explorer information includes the following sections:

- Signal Definition Explorer Windows
- Using the Signal Definition Explorer

Related Topics

- [Using the ESA Toolkit](#)



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Using the Signal Definition Explorer

Using the Signal Definition Explorer
SDE provides operations for many different library and file management and file editing tasks. This section includes following topics:

- [Overview: Using SDE and TSD Files](#)
- [Starting SDE](#)
- [Configuring the SDE Search Path](#)
- [Selecting TSD Files](#)
- [Editing TSD Files](#)
- [Operating on Multiple TSD Files](#)
- [Managing TSD Files](#)
- [Viewing TSD File Sample Data](#)

Related Topics:

- [About the Embedded Signal Analyzer Toolkit](#)

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Using the Signal Definition Explorer : Overview: Using SDE and TSD Files

Overview: Using SDE and TSD Files

Before you can use TSD files, one or more folders with TSD files must be installed on your workstation computer, tester computer, or network.

To use TSD files in a test program, follow these basic steps:

1. Start SDE (see Starting SDE).
2. If necessary for you to access the files you need, configure the search path to access libraries of available files (see Configuring the SDE Search Path).

Files are organized under wireless groups and standards. See Signal Catalog for descriptions of the file name formats.

3. To narrow down the number of TSD files to consider, apply a filter to find files for a wireless standard or instrument setting (see Selecting Files Using a Filter).

To find measurement (capture) TSD files, create a filter with the Measurement property, with value set to <Present>.

4. Read the Signal Catalog to understand how TSD files relate to test or measurement type.

5. View properties of a TSD file using the SDE editor to evaluate the suitability of that file for a test. General comments are in the Description field, in the Description & Notices tab.

6. For a source file, open a viewing tool to view data (see Viewing TSD File Sample Data).

7.

If a file meets your requirements, copy the file to a folder accessible to the test program (see [Copying a Single TSD File](#)). In addition to the TSD file, SDE copies any setup or utility files.
8.

If you have an instrument that requires separate I and Q data, use SDE to split the IQ data into separate I and Q TSD files (see [Splitting a TSD File Containing IQ Data Into Separate TSD Files](#)).
9.

Repeat steps 2 - 8 until all TSD files required by your program are copied to your development folders.

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Using the Signal Definition Explorer : Starting SDE

Starting SDE

SDE is a standalone tool installed with IG-XL, but you must start SDE from outside IG-XL using one of the following methods:

- To start SDE from the Windows start menu, select Start>Programs>Teradyne IG-XL Vxx.yy.zz_uflx>Tools>Signal Definition Explorer.
- To start SDE from the \bin folder of your IG-XL installation, double-click SignalDefinitionExplorer.exe.

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Using the Signal Definition Explorer : Configuring the SDE Search Path

Configuring the SDE Search Path

The SDE search path sets the library file folders that SDE searches when it is comparing TSD files to a filter.

The active library is the included folders in both the System TSD and User TSD libraries. The SDE filter searches the active library when looking for files that match the filter. SDE does not search excluded folders.

To configure the SDE search path:

1.

Select File>Configure Search Path.

The Configure Signal Library Search Path window appears.

- To remove one or more folders in the User TSD library, right-click the folder in the User TSD Library and select Remove Folders(s).

SDE removes the folder.

- To add one or more folders to the User TSD library, right-click a folder in the User TSD Library and select Add Folder(s).

The Browse For Folder window appears. Navigate to and select an existing folder or make a new folder. Click OK to add the folder to the SDE search path.

- To exclude one or more folders from either the System TSD library or the User TSD library, right-click a folder in the TSD Library and select Exclude Folder(s).

SDE dims the folder and subfolders and excludes them from the search path.

- To include an excluded folder in either the System TSD library or the User TSD library, right-click an excluded (dimmed) folder and select Include Folder(s).

SDE un-dims folder and subfolders and restores them to the search path.

- To include only one folder in the search path and exclude all others, right-click the folder and select Include this folder only.

2.

Click OK to close the window.

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Using the Signal Definition Explorer : Selecting TSD Files

Selecting TSD Files

You can select TSD files using these methods:

- [Configure a search path of folders for SDE to filter using TSD file properties you specify.](#)
- [Select one or more TSD files in the Signal Library view tree.](#)

You can select TSD files from either the system TSD library supplied by Teradyne (installed in the SignalLib folder) or the user TSD library. You specify the folders searched by SDE by configuring the search path.

Before you modify a TSD file from the system TSD library, copy it to a user folder. This allows you to revert to the original Teradyne-supplied file without reinstalling the entire library.

You can add or remove folders from the user library. You can enable or disable folders in either the system or user library. When a folder is disabled, SDE will not search the folder or show the files in the folder in the file tree view. The active library is the set of enabled folders.

See the following topics:

- [Selecting Files Manually](#)
- [Selecting Files Using a Filter](#)
- [Saving a Filter](#)
- [Adding a Filter](#)

Selecting Files Manually

You can select one or more TSD files using standard Windows multiselect commands in the Signal files view tree:

- To select a file, click the file name.

The selected file is highlighted.

- To select an additional file, Ctrl+click the file name.

The additional file is also selected and highlighted.

- To select a group (range) of files, click the file name that begins the range and then Shift+click the file name that ends the range.

The entire range of files, including the first and last file, is selected and highlighted.

You can also Ctrl+click to deselect individual files and Shift+click to deselect ranges.

Selecting Files Using a Filter

The Signal files tree shows all the TSD files in the active library. You can select files by setting up a filter to select only the files which match the conditions set in the filter.

To set up a TSD file section filter:

1. Click Select Filter Properties.

The Select Filter Properties window appears with a checklist of the properties for any currently selected files. If no files are selected, all properties of the active library files are listed. The box is checked for any properties currently in the file selection filter.

2. Click checkboxes for individual properties to include or remove the properties from the list to add to the filter.

Note: An easy way to filter for measurement (capture) TSD files is to clear the filter and add the Measurement property and set the value to <Present>.

3. Click OK.

The window closes. The checked properties are added to the File selection filter in the main window. Properties with a value you previously defined retain the value. Properties which were just added have the value <Any>.

The TSD files matching the properties are selected (highlighted) in the Signal files tree.

4. For a property in the filter:

a. Click a property value in the filter.

A drop-down list appears.

b. Select a value from the list or enter a value in the text box at the end of the list.

The value you selected or entered appears in the filter. SDE updates the Signal files tree and selects (highlights) only the files meeting the new filter criteria.

5. For a property value in the filter that you want to have a comparator other than "=" (default):

a. Click the comparator symbol between the property name and value.

A drop-down list appears.

b. Click the comparator you choose.

The comparator you selected appears in the filter. SDE updates the Signal files tree and selects (highlights) only the files meeting the filter criteria.

6. Repeat the steps above until you have the TSD file set you want.

You can save a file selection filter or add one or more saved file selection filters to the currently active filter. See [Saving a Filter](#) and [Adding a Filter](#).

[Saving a Filter](#)

After you set up a file selection filter, you can save the filter for later use. SDE filter files have the extension .tsf.

To save an SDE file selection filter:

1. Click Save Filter in the SDE main window.

The Save Current Filter window opens.

2. Select the folder where you want to save the SDE filter file.

3. Enter a file name for the filter file and click Save.

SDE saves the filter file to the folder you selected.

Adding a Filter

You can add a filter that was previously saved in a filter (.tsf) file to the currently active file selection filter.

To add a file selection filter in an SDE filter file:

1. Click Save Filter in the SDE main window.

The Add Filter window opens.

2. Navigate to a folder containing filter files, select a filter file and click Open.

The Add Filter window disappears. SDE adds the properties in the filter file to the active file selection filter.

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Using the Signal Definition Explorer : Editing TSD Files

Editing TSD Files

You can view and edit a TSD file using the SDE edit window. Before you can edit a file you must select it (see [Selecting TSD Files](#)) and open it. SDE allows you edit more than one TSD file, each with a separate SDE editor window.

Before you modify a TSD file from the system TSD library, copy it to a user folder. This allows you to revert to the original Teradyne-supplied file without reinstalling the entire library.

To edit a TSD file:

1. [Click the file name in the Signal files tree.](#)

[The file name is selected and highlighted.](#)

2. [Select File>Open Selected File or type Ctrl+O.](#)

[After SDE verifies the TSD file format, the SDE editor window opens with the Properties tab selected.](#)

[You can also select and open a file by double-clicking it.](#)

3. [Modify the TSD file in one or more of the ways described in the topics below.](#)

4. [Save the changes in the original file or a new file.](#)

5. [Click OK.](#)

[The SDE edit window closes.](#)

The following topics describe ways to edit/modify a file:

- [Adding a Property](#)
- [Removing a Property](#)
- [Changing a Property](#)
- [Changing a Property Value](#)
- [Adding a Notice](#)
- [Editing the Description](#)
- [Removing a Component](#)
- [Converting IQ Data to IF](#)
- [Importing a Signal Definition](#)
- [Importing a Signal Component](#)
- [Splitting a TSD File Containing IQ Data Into Separate TSD Files](#)

[Adding a Property](#)

To add a property to a TSD file:

1. [Open the file.](#)

After SDE verifies the TSD file format, the SDE editor window opens with the Properties tab selected.

2. [Select](#) Edit>Add Property or right-click in the Properties box and select Add Property on the context menu that appears.

A new property appears at the end of the list with *****New Property***** as the new property name and the property value empty.

3. Select a new name for the property from the property name drop-down list.

You can navigate to a set of properties beginning with a specific letter by typing the letter.

4. Enter a property value or select a new value for the property from the property value drop-down list.

The Properties list shows the new property and value. They are in the file the next time you save it. To operate on multiple files, see [Adding a Property To Multiple TSD Files](#).

Removing a Property

To remove a property:

1. Open the file.

After SDE verifies the TSD file format, the SDE editor window opens with the Properties tab selected.

2. Right-click the property you want to remove.

3. Select Remove Property.

The property disappears from the properties list and is not in the file the next time you save it. To operate on multiple files, see [Removing a Property from Multiple TSD Files](#).

Changing a Property

1. Open the file.

After SDE verifies the TSD file format, the SDE editor window opens with the Properties tab selected.

2. Select a new property from the property name drop-down list.

You can navigate to a set of properties beginning with a specific letter by typing the letter.

3. Enter a property value or select a new value for the property from the property value drop-down list.

The Properties list shows the new property and value. They are in the file the next time you save it.

Changing a Property Value

To change a property value:

1. Open the file.

After SDE verifies the TSD file format, the SDE editor window opens with the Properties tab selected.

2. Click the value box for the property and enter a property value or select a new value for the property from the property value drop-down list.

The new property value appears and is in the file the next time you save it.

Adding a Notice

To add a notice, such as copyright information:

1. Open the file.

After SDE verifies the TSD file format, the SDE editor window opens with the Properties tab selected.

2. Click the Description & Notices tab.

The Description and Notices boxes appear in the editor window.

3. Select Edit>Add Notice.

A new notice box appears.

4. Click the new notice text box and type the new notice.

The new notice is in the file the next time you save it.

Editing the Description

To edit the file description:

1. [Open the file.](#)

After SDE verifies the TSD file format, the SDE editor window opens with the Properties tab selected.

2. [Click the](#) Description & Notices tab.

The Description and Notices boxes appear in the editor window.

3. [Click in the description text box and type your changes.](#)

The new description is in the file the next time you save it.

Removing a Component

To remove a waveform component:

1. [Open the file.](#)

After SDE verifies the TSD file format, the SDE editor window opens with the Properties tab selected.

2. [Click the](#) Components tab.

The Components list appears in the editor window.

3. [Right-click the component you want to remove.](#)

The component disappears from the list and will not be in the file the next time you save it.

To operate on multiple files, see [Removing Components \(Sample Data\) From Multiple TSD Files.](#)

Converting IQ Data to IF

To convert IQ waveform data to IF data:

1. [Open the file.](#)

After SDE verifies the TSD file format, the SDE editor window opens with the Properties tab selected.

2. [Select](#) Tools>Convert IQ to IF.

SDE converts the IQ waveform data to IF. The IF data is in the file the next time you save it.

Importing a Signal Definition

To import a signal definition:

1. Open the file.

After SDE verifies the TSD file format, the SDE editor window opens with the Properties tab selected.

2. Select Tools>Import>Signal definition data.

The Import Signal Data Definition Data File browser opens and displays signal definition metadata (.meta) files.

3. Select a signal definition metadata file to import and click OK.

If the imported file contains any properties that duplicate ones in the TSD file, SDE opens the Handle Duplicates dialog.

- a. Select the duplicate behavior you want for each property and click OK.

SDE updates the editor display with the imported data.

Importing a Signal Component

To import a signal component:

1. Open a file.

After SDE verifies the TSD file format, the SDE editor window opens with the Properties tab selected.

2. Select Tools>Import>TSD file.

The Import TSD File browser opens.

3. Select a TSD file to import and click Open.

If the imported file contains any properties that duplicate ones in the TSD file, SDE opens the Handle Duplicates dialog.

a. Click Keep Old All.

The Handle Duplicates window closes.

SDE updates the editor display.

If the imported file contains more than one component, the Select Component dialog appears.

b. Select the component to import into the TSD file you are editing and click OK.

SDE adds the component you selected to the TSD file you are editing.

If the imported file contains only one component, SDE adds it to the TSD file you are editing.

Splitting a TSD File Containing IQ Data Into Separate TSD Files

SDE can create two TSD files, one with I data and the other with Q data, from a single TSD file containing IQ data. SDE can perform this operation on one or more TSD files.

To split one IQ TSD file into separate I and Q TSD files:

1. Right-click a TSD file in the Signal files view tree.

The context menu appears.

2. Select Split IQ Data into I and Q.

The Browse For Folder window appears.

3. Select a destination folder for the new files and click OK.

The browser window disappears. SDE creates a new TSD file with the same name as the input file with "_I" appended to the I data file and with "_Q" appended to the Q data file.

If the file contains MIMO data with multiple components, SDE splits each IQ component into separate "_In" and "_Qn" files, where "n" is the relative IQ component number in the original file.

To operate on multiple files, see Splitting Multiple TSD Files Containing IQ Data Into Separate TSD Files.



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Using the Signal Definition Explorer : Operating on Multiple TSD Files

Operating on Multiple TSD Files

SDE allows you to perform some editing operations on more than one file at once.
See the following topics:

- [Removing a Property from Multiple TSD Files](#)
- [Adding a Property To Multiple TSD Files](#)
- [Removing Components \(Sample Data\) From Multiple TSD Files](#)
- [Splitting Multiple TSD Files Containing IQ Data Into Separate TSD Files](#)

Removing a Property from Multiple TSD Files

You can remove a property from more than one TSD file at once. For example, you might use a set of files in a new program and with an instrument that does not need a property in the TSD files in the Teradyne Signals Library.

To remove a property from multiple TSD Files:

1. [Select one or more TSD files that you want to remove a property from.](#)
2. [Select](#) Tools>Remove Property.

The Remove Property dialog appears.

3. Type the name of the property to remove, or select the name of the property from the drop-down list, and click OK.

The Browse For Folder window appears.

4. Select the destination folder for the modified files and click OK.

The Browse For Folder window disappears. SDE removes the property from the files and writes them to the destination folder.

Adding a Property To Multiple TSD Files

You can add a property to one or more TSD files at once. For example, you might use a set of files in a new program with an instrument that requires a property that is not in the TSD files in the Teradyne Signals Library.

☒ **Note:**

If one or more files already contain the property, the value is overwritten by the new value. To add a property to multiple TSD files:

1. Select one or more TSD files that you that you want to add a property to.
2. Select Tools>Add/Overwrite Property.

The Add/Overwrite Property dialog appears.

3. Type the name of the property to add or select the name of the property from the drop-down list.
4. Type the value for a property to add or to overwrite the value of an already existing property of the same name, or select the value of the property from the drop-down list, and click OK.

The Browse For Folder window appears.

5. Select the destination folder for the modified files and click OK.

The Browse For Folder window disappears. SDE adds the property to the files or overwrites the value of any property with the same name, and writes the modified files to the destination folder.

Removing Components (Sample Data) From Multiple TSD Files

You can remove the components (waveform samples) from multiple TSD files. For example, you can create measurement (capture) TSD files, which do not require samples, from source TSD files. TSD files without components are much smaller than files with components.

To remove components from multiple TSD files:

1. Select one or more TSD files that you that you want to remove samples from.

2. Select Tools>Remove Sample Data.

The Browse For Folder window appears.

3. Select the destination folder for the modified files and click OK.

The Browse For Folder window disappears. SDE removes the component or components containing the waveform samples from the files and writes the modified files to the destination folder.

Splitting Multiple TSD Files Containing IQ Data Into Separate TSD Files

To split one or more IQ TSD files into separate I and Q TSD files:

1. Select one or more TSD files containing IQ data.

2. Select Tools>Split IQ Data into I and Q.

The Browse For Folder window appears.

3. Select a destination folder for the new files and click OK.

The Browse For Folder window disappears. SDE creates new TSD files, each with the same name as the input file with "_I" appended to the I data file and with "_Q" appended to the Q data file, and writes them to the destination folder.

If the file contains MIMO data with multiple components, SDE splits each IQ component into separate "_In" and "_Qn" files, where "n" is the relative IQ component number in the original file.

			
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Using the Signal Definition Explorer : Managing TSD Files

Managing TSD Files

During program development, you usually choose one or more TSD files for one or more wireless standards from the Teradyne Signals Library. Typically, you copy these files from the Signals Library to the folder where you are developing your test program. In this folder, you either use the TSD files without modification or them to meet the requirements for testing the device.

When you move or copy a TSD file, SDE also copies any VSA setup or VSA utility files referred to by the TSD file. SDE does not move or delete setup and utility files in case they are used by other TSD files.

TSD files in the Signal Library are available for copying and editing. SDE prompts you to be sure before you overwrite a TSD file in the Signals Library you installed with the Modulation Toolkit. Save a copy of the file in another folder before you modify it. If you need to recover a TSD file in the Signals Library after you have modified it, you must reinstall the Signals Library.

This section contains the following topics:

- | | |
|---|--------------------------------|
| • | Moving and Renaming a TSD File |
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- | | |
|---|---------------------------|
| • | Copying a Single TSD File |
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| • | Copying Multiple TSD Files |
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| • | Deleting a Single TSD File |
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| • | Deleting Multiple TSD Files |
|---|-----------------------------|

Moving and Renaming a TSD File

You can move a TSD File from the Teradyne Signals Library or the User Library to the folder you use while you are developing your test program.

You can rename the TSD file when you move it. If you decide to overwrite an existing file with the same name, SDE moves the file being replaced to the Recycle Bin.

To move a TSD file from a library to a development folder:

1. Select the TSD file.

2. Select File>Move File.

The New TSD File Name window appears.

You can also right-click a file name and select Move File from the context menu.

3. Select a destination file folder.

4. To change the file name, enter a new file name, and click Save.

The New TSD File Name window disappears, unless this is a special case (see below).

SDE moves the TSD file and copies any setup and utility files referenced by the TSD file. (SDE does not move setup and utility files in case they are used by other TSD files.)

Some special cases can occur when you move a TSD file.

- If a file with the same name already exists in the destination folder, the File Already Exists dialog appears.

- a. Click Yes to overwrite the file.

The dialog disappears. SDE moves the existing file to the Recycle Bin. Then SDE moves the TSD file, and copies any setup and utility files referenced by the TSD file, from the original folder to the destination folder.

- If the destination folder is write-protected, a Windows dialog appears offering to move the file to the My Documents folder.

- a. Click No.

The New TSD File Name window reappears.

- b. Select a different folder or enter a new file name and click **Save**.

SDE moves the TSD file and copies any setup and utility files referenced by the TSD file.

- If you try to move a TSD file to the IG-XL\SignalLib folder (the default location of the Teradyne Signals Library files), the Are You Sure dialog appears.

- a. Click **Cancel**.

Move TSD files to a different folder to use or modify them.

Copying a Single TSD File

You can copy a single TSD file from the Teradyne Signals Library or the User Library to the folder you use while you are developing your test program.

If you decide to overwrite an existing file with the same name, SDE moves the file being replaced to the Recycle Bin.

To copy a TSD file:

1. Click a single file in the Signal file view tree and select **File>Copy File(s)**, or right-click a file and select **Copy File** on the context menu.

The Browse For Folder window appears.

2. Navigate to and select a destination folder, or make and then select a new destination folder, and click **OK**.

The browser window disappears. SDE copies the TSD file and copies any setup and utility files referenced by the TSD file.

Some special cases can occur when you copy one TSD file or more than one. See [Copying Multiple TSD Files](#).

Copying Multiple TSD Files

You can copy one or more TSD files from the Teradyne Signals Library or the User Library to the folder you use while you are developing your test program.

If you decide to overwrite an existing file or files with the same name, SDE moves the file being replaced to the Recycle Bin.

To copy one or more TSD files:

1. [Select one or more TSD files.](#)

2. [Select File>Copy File\(s\).](#)

The Browse For Folder window appears.

3. [Navigate to and select a destination folder, or make and then select a new destination folder, and click OK.](#)

The browser window disappears. SDE copies the TSD files and any setup and utility files referenced by the TSD files. The SDE status bar shows the number of files copied.

Some special cases can occur when you copy one or more TSD files:

- [If a file with the same name already exists in the destination folder, the File Already Exists dialog appears.](#)

- a. [Click Yes to overwrite the file.](#)

The dialog disappears. SDE moves the existing file to the Recycle Bin. Then SDE copies the TSD file, and any setup and utility files referenced by the TSD file, from the original folder to the destination folder.

- [If the destination folder is write-protected, a Windows dialog appears offering to copy the file to the My Documents folder.](#)

- a. [Click No.](#)

The Browse For Folder window reappears.

- b. [Select a different destination folder and click Save.](#)

SDE copies the TSD file and any setup and utility files referenced by the TSD file.

- [If you try to copy a TSD file to the IG-XL\SignalLib folder \(the default location of the Teradyne Signals Library files\), the Are You Sure dialog appears.](#)

- a. [Click Cancel.](#)

Copy TSD files to a different folder to use or modify them.

Deleting a Single TSD File

When you delete a file using SDE, SDE moves the file to the Recycle Bin. (SDE does not delete setup and utility files in case they are used by other TSD files.)

To delete a single TSD file:

1. Right-click a TSD file in the Signal files view tree and select.

The Are You Sure dialog appears. The dialog warns you if the file is in the Signals Library.

2. Click Yes to want to delete the file.

SDE moves the file to the Recycle Bin.

Deleting Multiple TSD Files

When you delete one or more files using SDE, SDE moves the file or files to the Recycle Bin. (SDE does not delete setup and utility files in case they are used by other TSD files.)

To delete one or more TSD files:

1. Select one or more files to delete.

The Are You Sure dialog appears. The dialog warns you if the file is in the Signals Library.

- a. Click Yes if the files are not in the Signals Library and you want to delete the file or files.

SDE moves the file or files to the Recycle Bin.

- b. Click Cancel if the file or files are in the Signals Library.

Although the TSD files in the Signals Library are not protected, you can only recover them after you delete them by reinstalling the Modulation Toolkit, or by manually retrieving files from the Recycle Bin and restoring them to the correct folder in the Signal Library.

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SignalDefExplorerUsing.3.9



Using the Signal Definition Explorer : Viewing TSD File Sample Data

Viewing TSD File Sample Data

You can view waveform sample (component) data in a TSD file you select in SDE. The viewer tools available in SDE are:

- WaveScope
- VectorScope
- VSA

You can launch a viewer tool only if one and only one TSD file is selected. If more than one TSD file is selected, The View>ViewSignal menu item and the view icons are disabled.

To view sample data using WaveScope:

- Select a TSD file.
- Select View>View Signal>WaveScope or click the WaveScope icon on the toolbar.

WaveScope displays the samples in one or more separate WaveScope windows.

To view sample data using VectorScope:

- Select a TSD file.
- Select View>View Signal>VectorScope or click the VectorScope icon on the toolbar.

VectorScope displays the samples in a separate VectorScope II window.
To view sample data using VSA:

- | | |
|----|---|
| 1. | Select a TSD file. |
| 2. | Select View>View Signal>VSA or click the VSA icon on the toolbar. |

VSA displays the samples in a separate VSA window.
The View>ViewSignal>VSA menu item and the VSA icon are disabled under the following conditions:

- | | |
|---|--|
| • | VSA is not installed. |
| • | VSA does not have a valid license. |
| • | The TSD file does not have sample components, such as a capture file. |
| • | The TSD file does not have a VSA Setup property specifying a setup file. |
| • | Multiple TSD files are selected. |

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SignalDefExplorerWindows.2.1



Signal Definition Explorer Windows

Signal Definition Explorer Windows

All functions of the Signal Definition Explorer are available from two windows:

- [Signal Definition Explorer \(main\) window](#)
- [Signal Definition Explorer editor window](#)

The following topics describe the SDE windows, commands and toolbars:

- [Signal Definition Explorer Window](#)
- [File Selection Filter Controls](#)
- [SDE Commands and Toolbar](#)
- [SDE Editor Window](#)
- [SDE Editor Commands and Toolbar](#)



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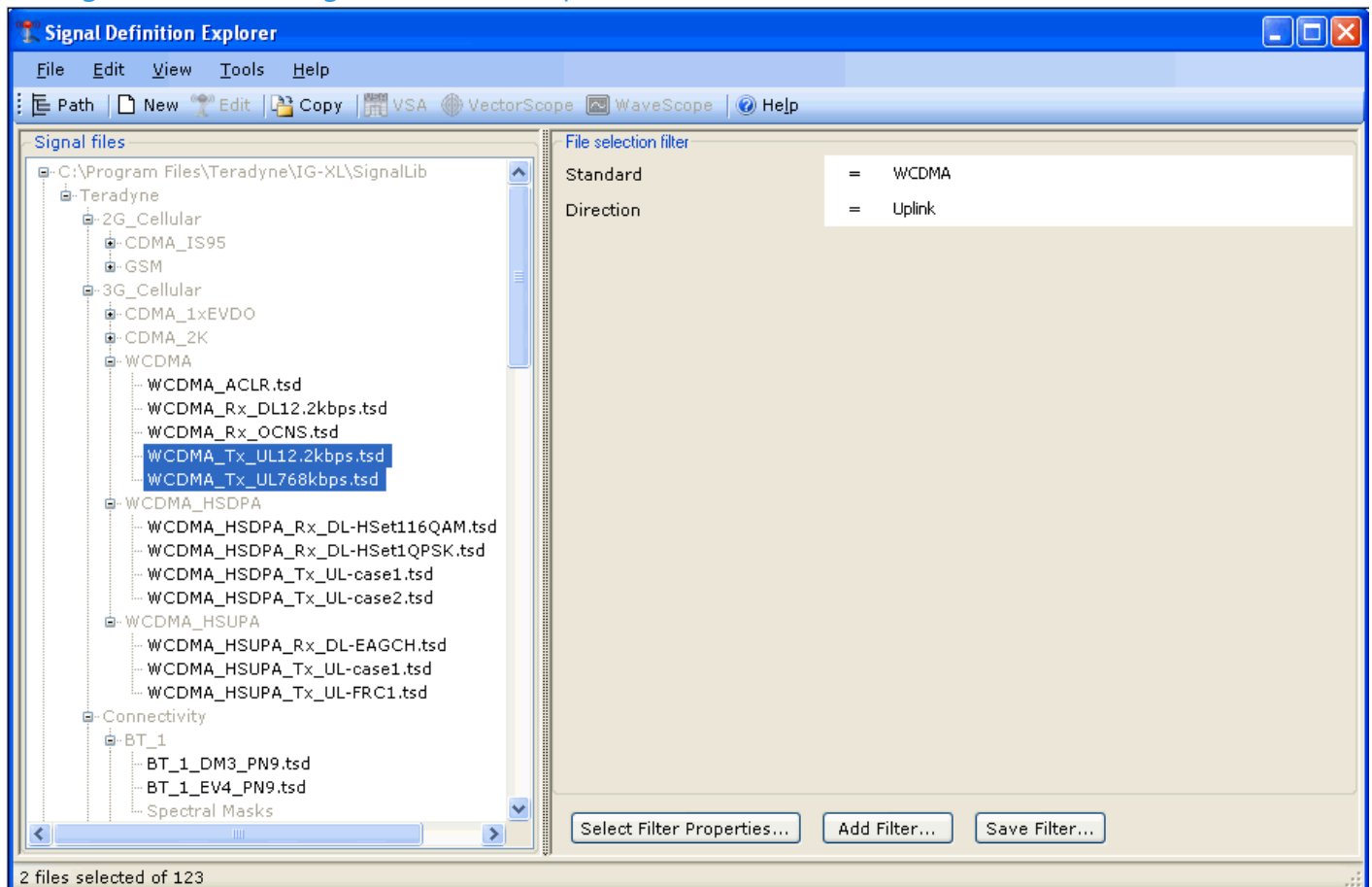
SignalDefExplorerWindows.2.2



[Signal Definition Explorer Windows](#) : Signal Definition Explorer Window

Signal Definition Explorer Window

The figure shows the Signal Definition Explorer main window.



Signal Definition Explorer Main Window

In addition to a menu bar and toolbar, the SDE window has the following parts:

- [File View Tree](#)

	File Selection Filter
	Status Bar

File View Tree

This tree on the left side of the window displays all TSD files in the Signal library, organized by folder. It includes folders and files in the Teradyne-supplied Signal Library (SignalLib folder) and any folders and files in a user library created by you or others.

The file tree is interactive: you select an individual file in the Signal files tree by clicking the file name. You can also select and deselect multiple files using the standard Windows multiselect commands: Ctrl+click for individual files and Shift+click for ranges.

You can also select files using a file selection filter, as described below. The figure above shows two files selected by the filter on the right side.

You can collapse or expand library folders. If a collapsed folder contains a file selected by a selection filter, the folder is highlighted in a lighter color than selected files.

Using an SDE command, you can exclude a folder of TSD files from the active library. Only the folders in the active library are checked when you select files with the filter.

File Selection Filter

This filter on the right side of the window displays a set of TSD file selection filter criteria. When the filter is applied, SDE selects and highlights in the view tree all TSD files that match the filter settings. The filter is applied to the active library (only included but not excluded folders), including both the Teradyne Signal Library and the user library.

The figure above shows a file selection filter with two TSD file properties: Standard and Direction. SDE used that filter to select the two files in the file view tree.

A selection filter consists of one or more TSD file properties, a comparator, and a property value. For more information, see File Selection Filter Controls.

Status Bar

The status bar at the bottom of the window displays messages when SDE completes an operation. In the figure above, the status message shows that the file selection filter found 2 TSD files meeting the filter criteria out of 123 files in the Signal (TSD) file libraries.

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SignalDefExplorerWindows.2.3

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Signal Definition Explorer Windows : File Selection Filter Controls

File Selection Filter Controls

An SDE file selection filter is a list of TSD file properties, each with a value and a comparator that defines how SDE will use the property and value as it selects TSD files that match the filter. You choose the properties to add to a filter using controls on the Select Filter Properties Dialog. See [Select Filter Properties Dialog Controls](#). You set property values and comparators using the comparator and property value controls, as described in the table. The SDE file selection filter has the following controls:

property list	Displays the list of properties in the current file selection filter. If all properties have been cleared, the list is empty.
property name	Displays the name of a property on the filter property list. You can only delete or add a property name using the Select Filter Properties dialog.
comparator (symbol between property and value)	Displays and sets how SDE uses a property and value as it selects files. Click the comparator to select one from a drop-down list. For numeric property values, the comparators are: = - equals != - not equal to < - less than

- `<=` - less than or equal to
- `>` - greater than
- `>=` - greater than or equal to
- `%N` - modulo
- `[n-m]` - between (Ex. Sample Rate [n-m] 16384 and 100000)
- `{,,}` - property is found in list

For string property values, the comparators are:

- `=` - equals
- `!=` - not equal to
- `[]` - property contains string value
- `![ ]` - property does not contain string value
- `{,,}` - property is found in list

String comparisons are not case sensitive.

property value

Displays and sets the value of a property in the filter. Click the value to select one from a drop-down list or enter a value in the text box at the end of the drop-down list.

Values are listed in the following order:

- Values found in the selected files, followed by a horizontal bar
- Values found in nonselected files in the library, followed by a horizontal bar

- [<Any>](#) - matches any value or no value
- [<Present>](#) - matches any value, but the property must be present in the file to match
- [<Not Present>](#) - matches any file in which the property is not present

Select Filter Properties	Opens the Select Filter Properties dialog. Use this dialog to define a filter. See Select Filter Properties Dialog Controls.
Add Filter	Adds a filter (properties, comparators, and values) from an SDE filter (.tsf) file.
Save Filter	Saves the current file selection filter to an SDE filter (.tsf) file.

		
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SignalDefExplorerWindows.2.4



Signal Definition Explorer Windows : File Selection Filter Controls : Select Filter Properties Dialog Controls

Select Filter Properties Dialog Controls

The Select Filter Properties button on the File selection filter opens the Select Filter Properties dialog. The dialog opens with a list of the properties in the currently selected TSD files. Any properties already in the filter are checked.

The Select Filter Properties dialog has the following controls:

check box	Selects and indicates whether a property will be added to the filter. Click the box to select or deselect the property. A check mark indicates the property is selected.
Show All	Shows all properties in files in the active library. This button is only shown when the properties from a selected set of files are shown.
Show Selected	Shows only the properties in selected files in the active library. This button is only shown when the properties from all files are shown.
Clear All	Deselects (unchecks) all properties shown. When you click OK, SDE clears the selection filter.
OK	Applies the selection filter. SDE updates the view tree to show the TSD files selected by the filter. SDE highlights selected files and highlights any collapsed folders containing selected files with a lighter color.
Cancel	Closes the Select Filter Properties dialog without modifying the selected properties in the filter.

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SignalDefExplorerWindows.2.5



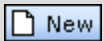
Signal Definition Explorer Windows : SDE Commands and Toolbar

SDE Commands and Toolbar

The SDE main window has the following menus for SDE commands:

- File Menu
- Edit Menu
- View Menu
- Tools Menu
- Help Menu

SDE has a toolbar with buttons (icon and label) for some commands. The following are the toolbar buttons and the corresponding menu commands:

 Path	File>Configure Search Path
 New	File>New
 Edit	File>Open Selected File
 Copy	File>Copy File(s)
 VSA	View>View Signal>VSA
 VectorScope	View>View Signal>VectorScope



View>View Signal>WaveScope



Help>Signal Definition Explorer Help (F1)

File Menu

File menu commands are:

New	New (Ctrl+N) Opens a new file in the SDE editor to allow you to create a new TSD file. See SDE Editor Window.
Open Selected File	Edit (Ctrl+O, double-click) Opens a TSD file in the SDE editor for editing. This command is disabled unless only one TSD file is selected in the view tree.
Open Signal Definition Data	Opens a dialog to import an ASCII metadata file produced by the EDA program which created waveform component data. This command is used only by design engineers creating TSD files.
Open Text File	Opens a text file containing waveform data such as a Gen4 waveform and displays it in the SDE editor.
Move File	Moves a TSD file from one folder to another and optionally renames the file. You can also rename a file in the current folder. You must have write access to both the source and destination folders.
Copy File(s)	Copy Copies one or more TSD files from one folder to another. You must have write access to the destination folder.
Delete File(s)	Deletes one or more TSDs file from a folder. (Del) You must have write access to the folder. SDE does not delete any referenced setup or utility files because they may be used by one or more other TSD files.
Configure Search Path	Path Opens the Configure Signal Library Search Path dialog. The dialog shows the system TSD library (Teradyne SignalLib) and the user TSD library in separate collapsible/expandable view trees. Use the view trees and context menu commands to include folders in the search

path used by the SDE filter. The active library is the folders included in the search path.

Right-click to view the context menu and select the following commands:

- **Include folder(s)**: Includes one or more folders in the search path.
- **Exclude folder(s)**: Excludes one or more folders from the search path.
- **Include this folder only**: Includes only the one selected folder in the search path.
- **Expand all**: Expands all folders in the view tree.
- **Collapse all**: Collapses all folders in the view tree.
- **Reset path**: Resets the search path.

In the user TSD library view tree only, select the following commands on the context menu:

- **Add folder**: Adds a folder to the library.
- **Remove folder(s)**: Removes one or more folders from the library.

Exit

Closes SDE. (Alt+F4)

Edit Menu

Edit menu commands are:

Select All

Selects all TSD files in the view tree. (Ctrl+A)

View Menu

View menu commands are:

View Signal

Accesses a submenu with the following signal viewer tools:



- **VSA**: Opens VSA for the selected TSD file. Enabled only if VSA is installed and licensed, and the TSD file has a setup file.



- **VectorScope**: Opens VectorScope for the selected TSD file.



- **WaveScope**: Opens one or more WaveScope windows for the waveform components in the selected TSD file.

This submenu is disabled if no files are selected or more than one file is selected.

Apply Filter

Applies the filter to the active library and selects the TSD files that match the filter criteria.

Expand All

Expands all folders in the Signal files view tree.

Collapse All

Collapses all folders in the Signal files view tree.

Tools Menu

Tools menu commands are:

Create TSD file(s) from signal definition data

Creates one or more TSD files and converts signal data from an EDA tool to TSD file format.

This command is used only by design engineers creating TSD files.

Create TSD file(s) from Text (Gen4) data

Creates one or more TSD files and converts the signal data from Gen4 format in a text file to TSD file format.

Split IQ data into I and Q

Splits a TSD file containing IQ waveform data into two separate TSD files, one containing I data and one containing Q data, and puts the modified files in a folder you specify.

	SDE names the new TSD files by appending “_I” or “_Q” to the file name of the input IQ TSD file. If the file contains MIMO data with multiple components, SDE splits each IQ component into separate “_In” and “_Qn” files, where “n” is the relative IQ component number in the original file.
Remove Property	Removes a property from one or more selected TSD files and puts the modified files in a folder you specify.
Add/Overwrite Property	Adds or overwrites a property in one or more selected TSD files and puts the modified files in a folder you specify.
Remove Sample Data	Removes sample data from one or more TSD files and puts the modified files in a folder you specify.
Check File Syntax	Checks the syntax of one or more TSD files and displays the result in the status bar.

Help Menu

Help menu commands are:

Signal Definition Explorer Help	 Help (F1) Opens the Signal Definition Explorer Window topic in UltraFLEX User Manuals.
ESA Toolkit Help	Opens About the Embedded Signal Analyzer Toolkit under UltraFLEX User Manuals>IG-XL Tools>Embedded Signal Analyzer (ESA) Toolkit in the IG-XL User Manuals available in IG-XL Help.
About Signal Definition Explorer	Opens an About window with information about Signal Definition Explorer software, including software version.

SignalDefExplorerWindows.2.6



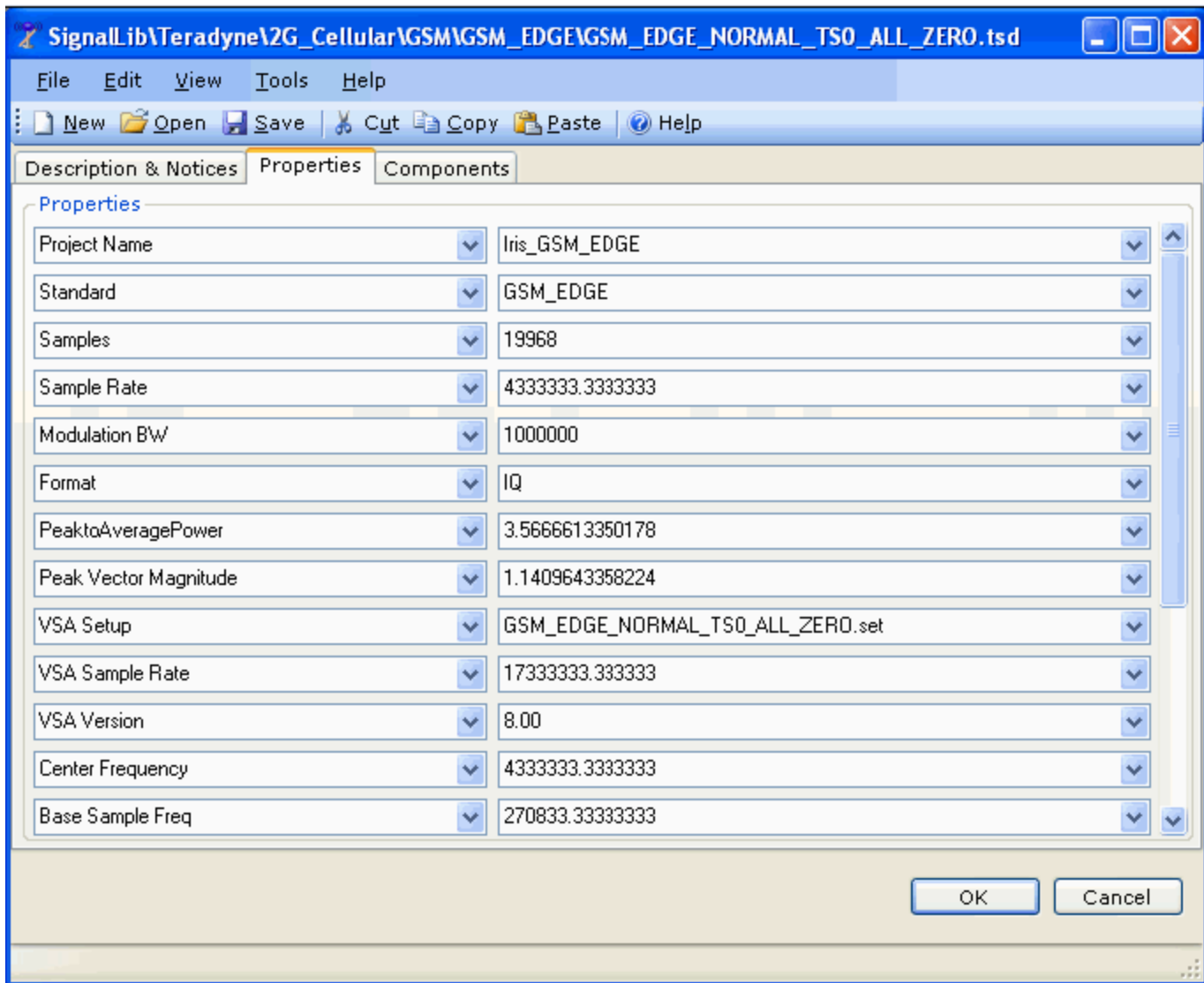
Signal Definition Explorer Windows : SDE Editor Window

SDE Editor Window

You use the SDE editor for creating and modifying TSD files for wireless standards, tester instruments, and tests for specific devices.

The SDE editor window is nonmodal. This means that you can use the main window and more than one editor windows at the same time. Each editor window operates on one TSD file.

The figure shows the SDE editor window. The editor window appears with name of the TSD file in the title bar and the Properties tab selected.



SDE Editor Window

The SDE editor has two buttons:

OK

Saves any changes you made to the TSD file and closes the editor window. If a file would be deleted or overwritten, SDE prompts you to make sure you want to complete the command.

Cancel

Closes the editor window without saving any changes you made.

In addition to menus, a toolbar, and buttons, the SDE editor window has the following tabs displaying data the in a TSD file open in the editor:

Description & Notices Displays two sets of information for a TSD file:

- An editable description of the TSD file, including source file information for Teradyne TSD files. Type in the description text box

to add or change information.

- A list of notices, including a copyright notice for Teradyne files. Right-click in the Notices pane and select a command to add one or more additional notices or notes, or to remove an existing notice. Type new information in a new notice text box.

Click OK to save the changes and close the editor window.

Properties

Displays a scrollable list of the properties and property values in a TSD file in two columns of text boxes with drop-down menus. SDE populates the properties menu with the names of all the properties in the active library. SDE also populates the value menu with all the values for the property in the active library. Type a property name or value in the text box or select a property or value from the drop-down menu.

Click OK to save the changes and close the editor window.

Components

Displays the signal components (waveform sample data) in a TSD file. Each component has a name, a sample data format, and a sample size. You cannot edit sample size. For paired data, size is the number of pairs. For N pairs, there are 2 x N values. Capture (measurement) TSD files do not have components. The component name can be any string, but names are typically restricted to I, Q, IQ, and IF to identify the type of waveform. To remove a component, right-click the component name and select Remove Component. You cannot create a component using SDE. To change the waveform sample data storage format, click the drop-down menu and select the sample data format: Double, Float, or Text. Click OK to save the changes and close the editor window.

SignalDefExplorerWindows.2.7



Signal Definition Explorer Windows : SDE Editor Commands and Toolbar

SDE Editor Commands and Toolbar

The SDE editor window has the following menus for editing commands:

- File Menu
- Edit Menu
- View Menu
- Tools Menu
- Help Menu

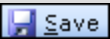
The SDE editor has a toolbar with buttons (icon and label) for some commands. The table lists the toolbar buttons and the corresponding menu commands.

 New	File> New
 Open	File> Open
 Save	File> Save
 Cut	Edit> Cut
 Copy	Edit> Copy
 Paste	Edit> Paste



File Menu

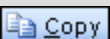
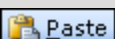
File menu commands are:

New	 (Ctrl+N) Resets the editor with a new, empty TSD file. SDE prompts you to save a file that is already open with unsaved edits.
Open	 (Ctrl+O) Opens an existing file selected using a file browser.
Save	 (Ctrl+S) Saves the file to the same file that you opened. If you create a file with the New command, Save operates as Save As. You must have write access to the file you are saving.
Save As	Opens a dialog allowing you to save the file to a new location with a new name. You must have write access to the folder you are saving the file to.
Close	Exits the editor window and saves any changes. The Close command is also performed by clicking OK.

Edit Menu

SDE saves changes when you select File>Close or click OK.

Edit menu commands are:

Cut	 (Ctrl+X) Removes selected text and places it on the clipboard.
Copy	 (Ctrl+C) Copies selected text and places it on the clipboard.
Paste	 (Ctrl+P) Pastes the text on the clipboard to the cursor position.

Add Property	Adds new property and value text boxes to the properties list on the Properties tab. The new property is named ***New Property***. Type a property name and value or select from the drop-down menus. This command is disabled unless you selected the Properties tab.
Add Notice	Adds a new notice text box to the Notices pane on the Description & Notices tab. The new text box is empty. Type the notice in the text box. This command is disabled unless you selected the Description & Notices tab.
Select All	Selects all text in the current control. (Ctrl+A) SDE highlights the selected text.

View Menu

View menu commands are:

Refresh	Refreshes the data in the editor window. (F5)
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Tools Menu

Tools menu commands are:

Import	Imports data from one of the following sources: Signal definition data: Imports signal definition data file from an EDA tool, converts the data to TSD format and merges the data with any data already in the editor. If the data contains duplicate properties, the Duplicate Behavior dialog appears prompting you to append the duplicate, keep the property and discard the duplicate, or replace the property with the duplicate from the imported file. Gen4 text file: Imports a waveform file in Gen4 format, converts the data to TSD format, and merges with the data in the editor.
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- [TSD File](#): Imports data from a TSD file and merges it with data currently in the editor.

If the data contains duplicate properties, the Duplicate Behavior dialog appears prompting you to append the duplicate, keep the property and discard the duplicate, or replace the property with the duplicate from the imported file. If the imported TSD file has more than one component, the Select Component dialog appears prompting you to select which component to add to the TSD file.

[Convert IQ to IF](#)

[Converts IQ data in a TSD file to IF data.](#)
The IQ data can be either separate I and Q components or a single IQ component. The single IF component replaces the IQ component or I and Q components.

[Help Menu](#)

[Help](#) menu commands are:

[Editor Help](#)



(F1)
[Opens the SDE Editor Window topic in UltraFLEX User Manuals.](#)

[Modulation Toolkit
Help](#)

[Opens the Modulation Toolkit topic in UltraFLEX User Manuals.](#)

			
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